

Townsend 5.1-5.3

Questions for Gallicchio

The spin bases looks entangled - we don't have every combination of particles in each spin state. Why?

In solving the energy eigenvalue problem, we have determined the eigenstates of the operator $2\hat{\mathbf{S}}_1 \cdot \hat{\mathbf{S}}_2$:

$$2\hat{\mathbf{S}}_1 \cdot \hat{\mathbf{S}}_2 \begin{bmatrix} |+\mathbf{z}, +\mathbf{z}\rangle \\ \frac{1}{\sqrt{2}}|+\mathbf{z}, -\mathbf{z}\rangle + \frac{1}{\sqrt{2}}|-\mathbf{z}, +\mathbf{z}\rangle \\ |-\mathbf{z}, -\mathbf{z}\rangle \end{bmatrix} = \frac{\hbar^2}{2} \begin{bmatrix} |+\mathbf{z}, +\mathbf{z}\rangle \\ \frac{1}{\sqrt{2}}|+\mathbf{z}, -\mathbf{z}\rangle + \frac{1}{\sqrt{2}}|-\mathbf{z}, +\mathbf{z}\rangle \\ |-\mathbf{z}, -\mathbf{z}\rangle \end{bmatrix} \quad (5.21)$$

and

5.1 Two spin $\frac{1}{2}$ particles

We could be nice and label by two particles in the state

$$|+z, +z\rangle$$

Or mean and label in any basis, such as

$$|+z, +x\rangle$$

They're complete basis, so there is a mapping between them.

$$|+x, +z\rangle = \frac{1}{\sqrt{2}}|+z, +z\rangle + \frac{1}{\sqrt{2}}|+z, -z\rangle$$

We can also rotate particles. Rotators applying to different particles commute, since they don't depend on order (unless entanglement?).

We define a new operation instead of the regular inner product and outer product: The direct product

$$|+z, -z\rangle = |+\mathbf{z}\rangle_1 \otimes |-\mathbf{z}\rangle_2$$

Which we can simplify notationally as just $|+\mathbf{z}\rangle_1 |-\mathbf{z}\rangle_2$.

5.2 Hyperfine splitting

$$\hat{H} = \frac{2A}{\hbar^2} \hat{\mathbf{S}}_1 \cdot \hat{\mathbf{S}}_2$$

Hydrogen microwave

5.3 The addition of Angular Momenta for two spin $\frac{1}{2}$ particles

Relative phase differentiates between

$$|+z, -z\rangle \text{ and } |-\mathbf{z}, +\mathbf{z}\rangle$$